V08: external pool measurement channels integrated into Simulink

8 scenarios × 168 h | 672 samples each | 900 s cadence | Class D evidence

1. Implemented time-series input

A new independent v08 model replaces the virtual sensor-delay blocks leading into the safety gate with two external temperature channels from independently validated CSV records. Metadata, UTC timestamps, sampling cadence and source provenance are checked first. The eight actual Simulink replays use previously generated SYNTHETIC v07 traces, not real facility measurements.

2. Thermodynamics and sensor logic

The gate now receives two externally supplied T1 and T2 signals; it does not directly access simulated true pool temperature. The first QA run detected an accidental second application of a sensor bias. It was repaired and all scenarios rerun. Thermodynamic and redundant cooling modules otherwise match v07. Shared-mode sensor errors remain intrinsically undetectable without an independent reference.

C_pool = rho V cp = 997 · 500 · 4182 = 2 084 727 000 J/K

T(k+1) = T(k) + 900 [Q_pool_physical - Q_loss]/C_pool

T1[k], T2[k] = imported calibrated-format CSV samples; no internal Ttrue input to gate

Q_DC = Q_HX_physical + Q_HP_source + Q_REJECT_main + Q_REJECT_backup + Q_unhandled

Q_HP_out = Q_HP_source + P_HP_compressor

Q_flow = rho (Vdot/3600) cp (T_supply - T_return)

u_rel(Q_flow) = sqrt[u_rel(Vdot)^2 + (sqrt(uT1^2+uT2^2)/deltaT)^2]

u(Q_from_pool_dT) = C_pool * sqrt(2) * u(T) / 900

3. Eight executed replay comparisons

Scenario	Max ΔT [°C]	Max ΔQ [W]	Latch Δ
Healthy	1.95e-08	0.0340	0
First valve	2.55e-08	0.0292	0
Both valves	3.01e-08	0.0435	0
Backup reject	1.95e-08	0.0340	0
Both rejects off	1.95e-08	0.0340	0
IT load shedding	1.95e-08	0.0340	0
Common sensor bias	1.95e-08	0.0340	0
Combined	2.92e-08	0.0418	0

4. Demonstration measurement uncertainty (assumptions only)

The calculation uses 30 m³/h, heat-exchanger delta-T of 3 K, independent temperature standard uncertainties of 0.2 K each and 5% flow uncertainty. These are assumed numbers, NOT calibrated instrument performance. Estimating 15-minute power directly from the 500 m³ pool water temperature is much noisier than flow calorimetry.

Q_flow = 104.236 kW; u(Q_flow) = 11.124 kW.

u(Q from 15-min whole-pool dT) = 655.2 kW.

5. Independent validation and next dependency

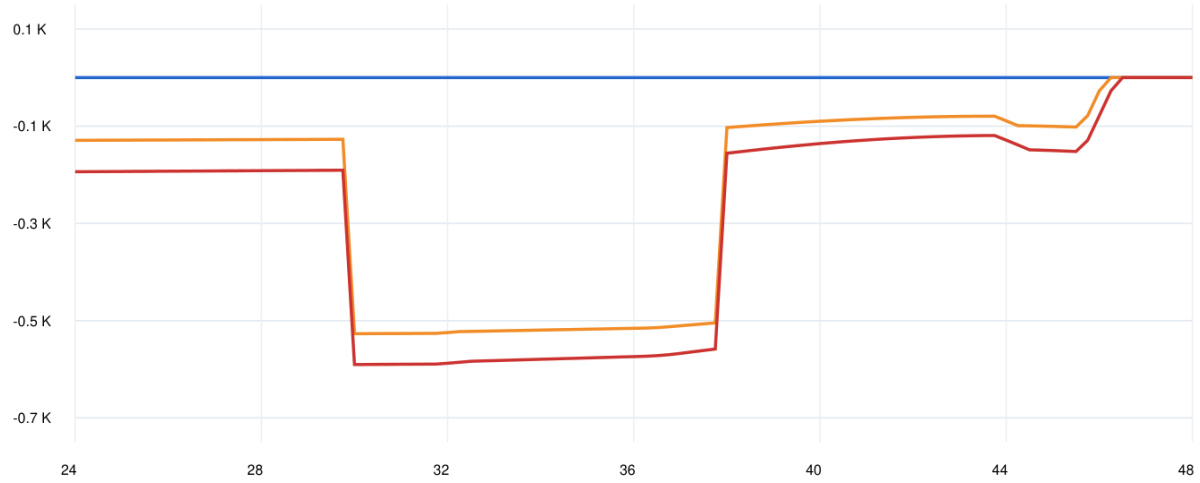
Independent Python checks passed 91565 conditions using native Simulink CSV exports and external sensor records. Provenance remains Class D. Physical Class A measurement streams with instrument IDs and calibration records are missing, so physical calibration, safety certification and plant performance estimates are NOT established.

Class A = measured; B = published study; C = manufacturer; D = synthetic / assumption.

6. Graphs from executed simulations

Odchylka externych snimacov v08 voci modelovanej vode

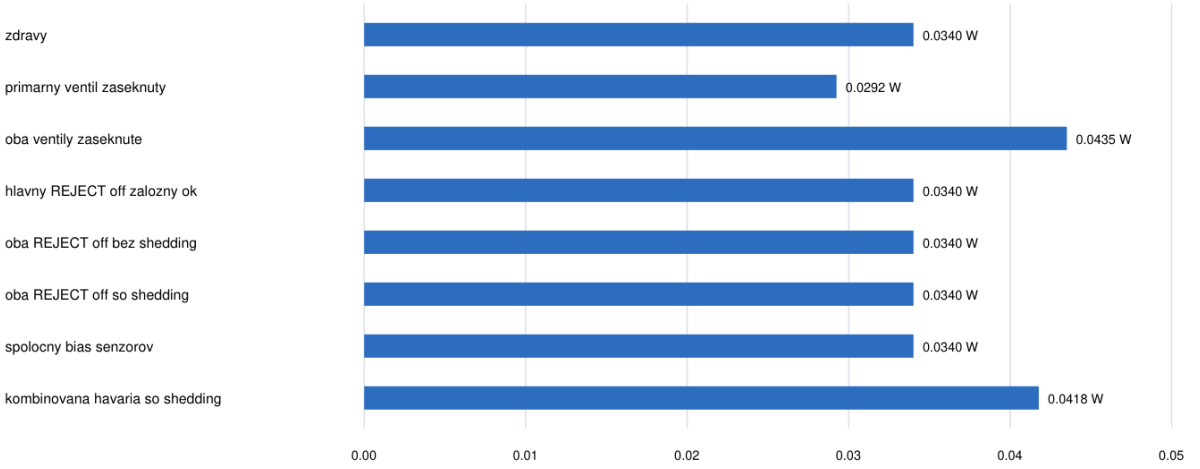
Modra = 0 K referencia, oranzova = T1-Tmodel, cervena = T2-Tmodel; spolocny bias -0.4 K



Cas [h], os Y = chyba T [K]; iba D-synteticky replay

Regresná zhoda v08 s v07 - maximálna absolútna odchýlka

Odchýlky Q_pool v watt-och pre rovnaké D-vstupy. 8 skutocnych Simulink replay behov.



Os X = maximalny rozdiel vypoctu tepla [W]